

## NIZHNY NOVGOROD STRIGINO, Russian Federation

WMO: 275530

Lat: 56.230N

Lon: 43.784E

Elev: 269

StdP: 14.55

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |       |       |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD<br>to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|-------|-------|-----|------|-----------------------|------|------|------|--------------------------|------|-------|
|               |            |      | 99.6%                         |     |       | 99%   |     |      | 0.4%                  |      | 1%   |      |                          |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB  | DP    | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                     | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)   | (g)   | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                      | (o)  | (p)   |
| 1             | -15.2      | -9.4 | -20.6                         | 1.8 | -14.9 | -14.8 | 2.5 | -8.7 | 18.7                  | 26.2 | 16.9 | 23.6 | 2.8                      | 310  | 0.552 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 19.5                   | 86.3            | 68.2 | 82.8 | 66.9 | 80.2 | 65.8 | 70.4                | 81.7 | 68.9 | 79.8 | 67.3 | 77.2 | 6.3                  | 210  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 66.5                            | 98.6 | 74.8 | 64.8 | 92.7 | 72.9 | 63.5 | 88.5 | 71.9 | 34.5          | 81.7 | 33.1 | 79.9 | 31.9 | 77.3 | 81.9              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |       |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |       |  |
|-------------------|------|------|----------------------------|-------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|-------|--|
| 1%                | 2.5% | 5%   | Mean                       |       | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |       |  |
|                   |      |      | Min                        | Max   | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |       |  |
| (a)               | (b)  | (c)  | (e)                        | (f)   | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |       |  |
| 16.4              | 14.3 | 12.5 | DB                         | -22.0 | 90.9               | 5.9 | 3.6                                                | -26.2 | 93.5       | -29.7 | 95.6       | -33.0 | 97.6       | -37.3 | 100.2 |  |
|                   |      |      | WB                         | -22.3 | 73.9               | 5.8 | 2.4                                                | -26.5 | 75.7       | -29.9 | 77.1       | -33.1 | 78.5       | -37.4 | 80.2  |  |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |
|------------------------------------------------------------------------|-------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                        |             | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 41.2        | 15.6  | 16.4  | 27.0  | 42.6  | 55.3  | 62.3  | 66.8  | 63.0  | 53.0  | 41.1  | 28.9  | 20.3  | (6)  |
|                                                                        | DBStd       | 20.50       | 12.78 | 12.85 | 9.05  | 9.16  | 8.91  | 7.60  | 6.38  | 7.10  | 7.51  | 8.09  | 9.72  | 12.00 | (7)  |
|                                                                        | HDD50       | 4927        | 1067  | 940   | 713   | 258   | 48    | 4     | 0     | 1     | 52    | 292   | 632   | 920   | (8)  |
|                                                                        | HDD65       | 8952        | 1532  | 1360  | 1178  | 671   | 318   | 140   | 56    | 123   | 367   | 740   | 1082  | 1385  | (9)  |
|                                                                        | CDD50       | 1706        | 0     | 0     | 0     | 37    | 213   | 373   | 520   | 405   | 141   | 17    | 0     | 0     | (10) |
|                                                                        | CDD65       | 256         | 0     | 0     | 0     | 0     | 19    | 59    | 111   | 62    | 5     | 0     | 0     | 0     | (11) |
|                                                                        | CDH74       | 2546        | 0     | 0     | 0     | 4     | 234   | 578   | 1078  | 608   | 44    | 0     | 0     | 0     | (12) |
|                                                                        | CDH80       | 717         | 0     | 0     | 0     | 0     | 50    | 153   | 316   | 194   | 4     | 0     | 0     | 0     | (13) |
| (14) Wind                                                              | WSAvg       | 5.7         | 6.3   | 6.3   | 6.4   | 6.0   | 5.7   | 5.1   | 4.5   | 4.5   | 4.7   | 5.7   | 6.3   | 6.4   | (14) |
| (15) Precipitation                                                     | PrecAvg     | 23.9        | 1.6   | 1.2   | 1.1   | 1.4   | 1.9   | 2.6   | 3.0   | 2.6   | 2.2   | 2.4   | 2.2   | 1.8   | (15) |
|                                                                        | PrecMax     | 32.4        | 3.2   | 3.1   | 3.4   | 3.4   | 5.7   | 5.6   | 8.1   | 6.4   | 5.6   | 6.5   | 3.9   | 3.6   | (16) |
|                                                                        | PrecMin     | 17.3        | 0.2   | 0.0   | 0.2   | 0.1   | 0.1   | 0.5   | 0.2   | 0.1   | 0.3   | 0.1   | 0.7   | 0.2   | (17) |
|                                                                        | PrecStd     | 4.1         | 0.7   | 0.6   | 0.7   | 0.7   | 1.3   | 1.2   | 1.7   | 1.3   | 1.1   | 1.5   | 0.8   | 0.8   | (18) |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB          | 37.7  | 38.5  | 52.8  | 73.5  | 85.5  | 89.4  | 92.9  | 91.8  | 79.0  | 66.4  | 52.0  | 42.9  | (19) |
|                                                                        |             | MCWB        | 35.9  | 35.4  | 42.9  | 55.4  | 65.5  | 67.4  | 68.6  | 68.6  | 62.9  | 55.0  | 48.5  | 41.8  | (20) |
|                                                                        | 2%          | DB          | 35.5  | 36.0  | 45.0  | 68.2  | 80.5  | 84.3  | 86.3  | 84.9  | 73.8  | 59.4  | 46.5  | 39.2  | (21) |
|                                                                        |             | MCWB        | 34.3  | 34.1  | 38.1  | 52.8  | 62.4  | 66.8  | 68.5  | 68.2  | 61.2  | 52.6  | 44.5  | 37.9  | (22) |
|                                                                        | 5%          | DB          | 33.9  | 34.5  | 41.3  | 63.0  | 75.5  | 80.5  | 83.2  | 80.4  | 69.9  | 55.6  | 43.3  | 35.9  | (23) |
|                                                                        |             | MCWB        | 32.9  | 33.1  | 36.0  | 50.3  | 59.9  | 65.7  | 67.9  | 66.2  | 59.2  | 50.7  | 41.6  | 34.6  | (24) |
|                                                                        | 10%         | DB          | 32.3  | 33.3  | 38.4  | 58.8  | 71.6  | 76.9  | 80.3  | 76.6  | 65.9  | 52.6  | 40.6  | 34.0  | (25) |
|                                                                        |             | MCWB        | 31.2  | 32.1  | 34.5  | 47.9  | 57.8  | 64.1  | 67.0  | 64.5  | 57.4  | 48.9  | 39.3  | 33.0  | (26) |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB          | 36.2  | 36.2  | 43.9  | 56.9  | 66.9  | 71.7  | 73.6  | 71.8  | 65.3  | 56.9  | 49.0  | 41.7  | (27) |
|                                                                        |             | MCDB        | 36.8  | 37.2  | 50.7  | 68.2  | 81.3  | 82.2  | 84.2  | 85.0  | 74.3  | 62.4  | 51.7  | 42.3  | (28) |
|                                                                        | 2%          | WB          | 34.1  | 34.5  | 39.2  | 54.2  | 64.1  | 69.3  | 70.9  | 69.5  | 62.5  | 53.8  | 44.9  | 37.7  | (29) |
|                                                                        |             | MCDB        | 34.7  | 35.4  | 43.4  | 65.5  | 76.5  | 81.0  | 81.2  | 81.5  | 71.1  | 58.2  | 46.0  | 38.5  | (30) |
|                                                                        | 5%          | WB          | 32.9  | 33.2  | 36.7  | 51.5  | 61.8  | 67.3  | 69.3  | 67.4  | 60.2  | 51.5  | 41.7  | 34.9  | (31) |
|                                                                        |             | MCDB        | 33.6  | 34.2  | 39.9  | 61.9  | 72.6  | 77.4  | 79.9  | 77.5  | 67.8  | 55.5  | 42.9  | 35.6  | (32) |
|                                                                        | 10%         | WB          | 31.2  | 31.9  | 35.0  | 48.9  | 59.5  | 65.3  | 67.9  | 65.5  | 58.1  | 49.2  | 38.8  | 33.0  | (33) |
|                                                                        |             | MCDB        | 32.3  | 33.0  | 37.7  | 57.5  | 68.8  | 74.2  | 78.0  | 74.3  | 64.9  | 52.2  | 39.9  | 33.8  | (34) |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR        | 10.9  | 13.4  | 14.3  | 17.5  | 21.0  | 19.6  | 19.5  | 18.9  | 16.3  | 11.3  | 8.2   | 9.3   | (35) |
|                                                                        |             | MCDBR       | 7.7   | 9.5   | 15.9  | 24.7  | 26.7  | 24.0  | 25.0  | 25.9  | 23.1  | 16.9  | 10.7  | 7.4   | (36) |
|                                                                        | 5% WB       | MCWBR       | 7.5   | 8.4   | 10.8  | 12.7  | 12.2  | 10.6  | 10.6  | 11.6  | 12.1  | 11.3  | 9.4   | 7.0   | (37) |
|                                                                        |             | MCWBR       | 8.0   | 9.3   | 13.3  | 22.9  | 23.6  | 21.4  | 21.6  | 23.4  | 20.9  | 15.8  | 9.9   | 7.2   | (38) |
| (40) Clear-Sky Solar Irradiance                                        | taub        |             | 0.287 | 0.322 | 0.355 | 0.409 | 0.386 | 0.380 | 0.408 | 0.428 | 0.374 | 0.356 | 0.314 | 0.285 | (40) |
|                                                                        |             | taud        | 2.085 | 2.046 | 2.041 | 2.123 | 2.332 | 2.377 | 2.316 | 2.241 | 2.371 | 2.313 | 2.203 | 2.063 | (41) |
|                                                                        | Ebn at Noon |             | 184   | 224   | 249   | 254   | 271   | 274   | 262   | 247   | 246   | 215   | 175   | 149   | (42) |
|                                                                        |             | Edh at Noon | 19    | 30    | 39    | 43    | 36    | 36    | 37    | 37    | 29    | 23    | 17    | 16    | (43) |
| (44) All-Sky Solar Radiation                                           | RadAvg      |             | 183   | 452   | 937   | 1264  | 1712  | 1762  | 1770  | 1401  | 850   | 388   | 150   | 99    | (44) |
|                                                                        | RadStd      |             | 34    | 91    | 108   | 131   | 113   | 161   | 169   | 148   | 123   | 67    | 28    | 26    | (45) |

## Historical Trends

|      |                        | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------|------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|      |                        |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
| (46) | Station Only           | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| (47) | Regional (6 neighbors) | +1.15 | N/A     | N/A    | N/A     | N/A   | N/A   | -243        | -324  | N/A   | N/A   |
|      |                        | +1.13 | N/A     | N/A    | N/A     | N/A   | N/A   | -238        | -322  | N/A   | N/A   |

Nomenclature: See separate page